

Predicto-Forex

Can a person at home find a way to profit from currency markets?

A plain-English explanation of what this project does, what it found, and what it cost to find out · Fifteen research sessions · July 2026

The one-paragraph version. We built software to test whether patterns in currency prices can be turned into profit. We tested a lot of ideas. Almost all of them failed — and we made very sure they had really failed, rather than us being bad at looking. One idea survived: currencies move in a small, consistent way each morning when London's banks open. Over twenty-one years of data that pattern is unmistakably real. Whether it makes money comes down to a single number we cannot measure from historical data, and measuring it is the next step.

1. What is forex, and what are we trying to do?

Forex (foreign exchange) is the market where currencies are traded. If you have ever changed money for a holiday, you have used it. Professionally, it is the largest financial market on earth — several trillion dollars change hands daily.

Prices move constantly. The euro might be worth \$1.08 this morning and \$1.084 this afternoon. Our question is simple to state and hard to answer:

Can we predict, better than a coin flip and by enough to cover our costs, whether a currency will rise or fall?

Note the two conditions. Beating a coin flip is not enough on its own. You must beat it by enough to pay for the privilege of trading — and that turns out to be the hard part.

Currency pair. Currencies are always priced against each other, never alone. `EURUSD` means "how many US dollars does one euro cost?" The first currency is the *base*, the second the *quote*. We study the seven most-traded pairs, all involving the US dollar.

Pip. The standard small unit of price movement — 0.0001 of the price for most pairs. If EURUSD goes from 1.0800 to 1.0805, it moved 5 pips. A typical day sees EURUSD move about 39 pips. Everything in this document is measured in pips.

Going long and going short. Buying something expecting it to rise is going *long*. Selling something you do not own, expecting to buy it back cheaper, is going *short*. Both are ordinary in currency markets, because buying one currency is automatically selling the other. Our strategy goes short.

2. The obstacle that shaped everything: the toll booth

Every time you trade you pay a small fee. It is not usually charged as a fee — it is built into the price. You buy at a slightly higher price than you can sell at, and the gap between those two prices is the **spread**.

Spread. The difference between the price to buy and the price to sell at the same instant. For EURUSD it is about 0.3 pips. You pay it on the way in and on the way out, so a complete trade (a *round trip*) costs roughly 0.3–0.8 pips depending on how well you execute.

That sounds tiny. It is tiny — about 1% of a typical day's movement. But you pay it on *every single trade*, and this arithmetic sank our first serious attempt:

What our first model did	Result
Predicted the next 15 minutes, traded constantly	—
Profit before costs, per year	+24.4%
Cost of all that trading, per year	–90.7%
Actual outcome	Large loss

The predictions were not worthless. They were just far too small to pay their own toll: the model predicted moves of about 0.17 pips while each trade cost about 1 pip. It was correct often enough to be interesting and nowhere near correct enough to be profitable.

This is the single most important lesson of the project. Every strategy we tested failed in the same way. The bottleneck was never "can we predict?" — it was always "can we predict by *enough*?" Trading costs are not a detail to add at the end. They decide the answer.

3. How do you know if you have found something real?

This is the part that consumed most of our effort, and it is worth explaining, because it is where nearly everyone who tries this goes wrong.

Financial data is extremely noisy. If you test enough ideas against it, some will look brilliant purely by chance. The danger is not failing to find a pattern — it is **finding one that isn't there and believing it**.

The coin-flip problem

Flip a coin ten times and get six heads. Is the coin biased? Your best guess is 60% heads. But the honest range of possibilities runs from about 26% to 88% — which includes 50%. You cannot tell this coin from a fair one. You need far more flips.

Confidence interval. The honest range around a measurement, rather than a single number. When we say "the range includes zero", we mean: *this might be real skill, or it might be luck, and this data cannot distinguish them*. Almost every result in this project looked like that until the very end.

Sharpe ratio. The standard measure of investment quality: profit relative to how wildly that profit swings. Roughly: 0 = you made nothing; 0.4 = about what a stock-market index fund gives you; 1.0 = genuinely good; 2.0+ = professional hedge fund. We agreed in advance that 0.5 was our minimum bar for taking anything seriously.

Four traps we actually fell into

These are not hypothetical textbook warnings. Each one produced a convincing, wrong result in this project before we caught it.

The trap	What happened to us
Seeing the future	Software that accidentally uses tomorrow's information to predict tomorrow looks spectacular and is worthless. We wrote tests that corrupt future data and confirm nothing in the past changes — then deliberately broke our own code to check the tests actually catch it.
Counting the same evidence repeatedly	One measure of trader positioning looked like our best-ever result. But we had counted 2,382 heavily overlapping observations as if independent, when there were really about 120. Corrected, the result vanished completely.
Using half the price	We initially used only the "sell" price rather than the midpoint. At quiet hours this manufactured a pattern that looked overwhelmingly strong and was pure measurement artifact.
Testing until something works	Test 50 ideas and 2–3 will look great by chance alone. We tracked every idea tested (~180 in total) and raised the bar for significance accordingly.

The general principle: every one of those errors made results look *better* than reality. Mistakes in this field are not random — they flatter you. That asymmetry is why so much of this project is defensive machinery rather than clever prediction.

The sealed envelope

The strongest protection is simple. We locked away several years of data and refused to look at it while experimenting. When we finally had a candidate worth testing, we wrote down our prediction in advance, committed it permanently, and then looked **once**.

Why once? Because if you peek, adjust, and re-test, you eventually find something that passes by luck. The software enforces this: it refuses to open the sealed data without a written prediction on file, and it logs every access permanently.

4. What we tried, and what happened

Over fifteen sessions we tested essentially every standard approach. Here is the honest scorecard.

Approach	What it means	Result
Momentum & trend	Assume recent movement continues	No signal
Mean reversion	Assume unusual moves snap back	No signal
Technical indicators	RSI, ATR and similar standard tools	No signal
Interest rates	Money flows toward higher-paying currencies	No signal
Currency strength	Watch seven pairs to separate "euro rose" from "dollar fell"	No signal
Trader positioning	Official data on what speculators are betting	Artifact — vanished when corrected
Market conditions	Do signals work in calm markets but not turbulent ones?	Refuted — 0 of 7 tests
Machine learning	Calibration, model ensembles	Made results worse
Time-of-day patterns	Do currencies behave differently at certain hours?	SURVIVED

The clearest single result: we screened 82 different predictive features. **One** passed even a basic significance check — where pure chance would produce about four. We found *fewer* apparent patterns than randomness alone generates. That is strong evidence the patterns genuinely are not there.

An unexpected finding about complexity. We imported sophisticated techniques from a sibling sports-prediction project. They made results consistently worse. The constraint was never model sophistication — it was that there was nothing in the data to find. More complexity applied to no signal produces confident nonsense.

5. The one thing that survived

The survivor is not a price pattern at all. It is a **structural** fact about how the market is organised.

The finding: the US dollar tends to strengthen slightly during the London morning, when Europe's banks open and a large volume of business flows through the market at a predictable time each day. The trade: sell the euro against the dollar at 9am London time, buy it back four hours later. There is nothing to predict — you simply act at the same time every day.

This matters because it is a *reason*, not just a pattern. It exists because of how the trading day is structured, which is why it does not evaporate the way statistical flukes do.

Twenty-one years of evidence

We extended our data back to 2005 specifically to test it. The crucial row is the first: that decade played no part in discovering the effect, so it is genuinely independent evidence.

Period	Trades	Gain per trade	Chance of a fluke
2005–2014 — never examined while searching	2,975	1.40 pips	1 in 340
2015–2022 — where we found it	2,076	2.88 pips	< 1 in a million
2023–2026 — the sealed envelope	907	1.03 pips	1 in 11
All 21.5 years combined	5,958	1.86 pips	1 in a billion

Twenty of the twenty-two years were profitable. The effect survived the 2008 financial crisis, the rise of computerised trading, and a five-fold collapse in trading costs. It shows no sign of fading.

A lesson in honesty about numbers. Notice the middle row is roughly double the others. That is the period we searched to *find* the effect — and searching inflates whatever you find. The truthful estimate, excluding that window entirely, is **1.31 pips**. The number you discover something at is almost always better than the truth, and the gap here was 2.2×.

6. Does it actually make money?

The gain is 1.31 pips per trade. The cost is whatever it costs to execute. So everything depends on a comparison of two small numbers.

To measure costs properly, we reconstructed real trades from **tick data** — every individual price change, roughly 80,000 per day — to see exactly what price we would have got at the moment of trading.

Cost per round trip	Profit per trade	Sharpe ratio	Verdict
0.80 pips — our old assumption	0.51 pips	0.34	Below bar
0.35 pips — actually measured	0.97 pips	0.64	CLEARs BAR
0.55 pips — with modest broker costs	0.77 pips	0.51	CLEARs BAR
0.75 pips — with higher broker costs	0.57 pips	0.37	Below bar

We had been assuming costs **2.3× higher than reality**. Correcting that changes the conclusion: measured properly, the strategy clears our agreed quality bar, on data that played no part in finding it. That is the first time anything in this project has managed that.

7. What we still don't know

The remaining uncertainty is narrow, specific, and honestly stated.

Slippage. The gap between the price you see when you decide to trade and the price you actually get. Caused by delay, by your broker's markup, and by the market moving in the meantime. **It cannot be measured from historical data.** It requires actually placing orders.

The entire remaining question: our budget for slippage is about **0.18 pips per trade leg**. Below that, the strategy is worth trading. Above it, it is not. Everything else has been measured; this one number has not.

Three further honest caveats:

- **Broker choice may now matter more than everything in this repository.** Our 0.35 pip figure comes from a high-quality data feed. A retail broker adds their own markup, and that markup varies enormously between providers.
- **This is a known effect.** London-open patterns are documented publicly. Reassuring for validity — we are not imagining it — but it means we are rediscovering, not discovering, and known effects can attract enough traders to erode them.
- **The returns are modest.** Roughly 1–3% per year unleveraged. Real, but not life-changing, and the effort required is substantial.

What happens next

The next step is to place real trades at the smallest size the market allows — about 10 cents per pip — purely to measure slippage. A full year of daily trading at that size risks roughly \$20–30 in total. That is the cost of an experiment, not an investment.

Notably, a practice account *cannot* answer this: simulated accounts fill orders at the quoted price, which flatters precisely the number we need to measure.

8. What this project actually produced

Two things exist here, and it is worth being clear about which is which.

A research pipeline that reliably tells the truth. It downloads and validates data, prevents accidental cheating, models real trading costs, and applies statistical tests strong enough to destroy our own promising results — which it did, twice, to findings we were excited about. 200 automated tests protect it, and the two most critical safeguards have been verified by deliberately breaking the code to confirm the alarms sound.

One small, real, structural trading effect, established across 21 years and independently confirmed on data that played no role in finding it — whose profitability rests on a single unmeasured execution number.

The honest summary. Most of what we tested does not work, and we now know that rigorously rather than vaguely — which is itself a real result, and the opposite of what most people in this field conclude. One thing does work, modestly. Whether it is worth trading is now a practical question about execution, not a research question about prediction.

A note on method. Every number in this document is measured rather than assumed, and every result that changed our conclusions was verified by trying to break it first. Several findings we were initially excited about are recorded here as failures, because that is what they turned out to be. The full technical record, including every mistake and its correction, is in `CLAUDE.md` in this repository.